Math 4141: Geometry and Differential Calculus

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Math 4141: Geometry and Differential Calculus — Credit: 4.00

Course Contents:

2D Coordinate Geometry: Change of axes: Transform of coordinates, Simplification of equations of curves; Pair of straight lines: Homogeneous second-degree equations, Conditions for general second-degree equations to represent a pair of straight lines, Angle between the lines, Pair of straight lines joining the origin to the points of intersection of the curve and a line, Circles and system of circles: Tangent and normal, Radial axis and its properties, Parametric coordinates.

3D Coordinate Geometry: Rectangular coordinates, Direction cosines and angle between two lines, The plane and the straight lines, The equation of a sphere, The standard forms of equations of the central conicoid, cones and cylinders.

Differential Calculus: Limits, Continuity and Differentiability; Differentiation of explicit and implicit functions and parametric equations; Significance of derivatives, Differentials; Successive differentiation of various types of functions; Leibnitz's Theorem; Rolle's Theorem; Mean Value Theorem; Taylor's Theorem in finite and infinite forms; Maclaurin's Theorem in finite and infinite forms; Lagrange's form of remainders; Cauchy's form of remainders; Expansion of functions by differentiation and integration; Partial differentiation; Euler's Theorem; Tangent, maximum and minimum values of functions and points of inflection; Applications of Differential Calculus; Evaluation of indeterminate forms by L'Hospitals rule; Curvature, the center of curvature and chord of curvature; Evolutes and involutes; Asymptotes, Envelopes, Curve tracing.

Pair of straight Lines

Pair of straight Lines: $ax^2 + 2hxy + by^2 = 0$

Angle between the line: $\tan\theta = \frac{2\sqrt{h^2-ab}}{a+b}$

If pair of straight lines are perpendicular: $a + b = 0$

If pair of straight lines are parallel (coincide): $h^2 = ab$

The equation of a **general pair of straight lines** is the product of the equations of the two lines:

$(A_1x + B_1y + C_1)(A_2x + B_2y + C_2) = 0$. Each factor represents one line. Expanding gives a general second-degree equation:

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, with coefficients that make it factor into those two lines.

If both lines pass through the origin, then $C_1 = C_2 = 0$, so the equation is homogeneous:

$(A_1x + B_1y)(A_2x + B_2y) = 0$. For example, $(x + 3y - 1)(x + 3y + 5) = 0$ is a pair of distinct parallel lines.

Details on the angle between two lines and the slops

Let the two lines make angles α and β with the positive x-axis. Their slopes are

$$m_1 = \tan\alpha, \qquad m_2 = \tan\beta.$$

The angle between the lines is $\theta = |\alpha - \beta|$ (taking the smaller angle). The tangent subtraction identity says

$$\tan(\alpha - \beta) = \frac{\tan\alpha - \tan\beta}{1 + \tan\alpha\tan\beta}.$$

Replacing $\tan\alpha$ and $\tan\beta$ by the slopes gives

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|.$$

The absolute value ensures the angle is nonnegative. If $1 + m_1 m_2 = 0$, the lines are perpendicular, so $\theta = 90^\circ$ and $\tan\theta$ is undefined.

For the **homogeneous pair**

$$ax^2 + 2hxy + by^2 = 0,$$

write each line through the origin as $y = mx$. Substitution gives

$$bm^2 + 2hm + a = 0.$$

Its roots m_1, m_2 are the slopes of the lines. Using

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$$

and the sum and product of the roots gives

$$\boxed{\tan\theta = \frac{2\sqrt{h^2 - ab}}{|a + b|}} \quad \text{when } a + b \neq 0.$$

If $a + b = 0$, the lines are perpendicular, so $\theta = 90^\circ$. This applies to two distinct real lines, for which $h^2 > ab$.

General Equation of Second degree (Class work)

Q. Find the condition that the general equation of the 2nd degree $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ may represent a pair of straight lines. Find point of intersection.

Solution: Given $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$……..(1)

If the origin transferred to the point (α, β)

Put $x = \alpha + x'$

and $y = \beta + y'$

Form equation (1) we get $ax'^2 + 2hx'y' + by'^2 + 2(a\alpha + h\beta + g)x' + 2(h\alpha + b\beta + f)y' + a\alpha^2 + 2h\alpha\beta + b\beta^2 + 2g\alpha + 2f\beta + c = 0$……(2)

The equation (1) may represent a pair of straight lines, if it is reduced to a homogeneous equation in x and y. This is possible if the coefficient of x and y and the constant term equal to zero

$a\alpha + h\beta + g = 0$……..(3)

$h\alpha + b\beta + f = 0$……(4)

and $a\alpha^2 + 2h\alpha\beta + b\beta^2 + 2g\alpha + 2f\beta + c = 0$

$\Rightarrow a\alpha^2 + h\alpha\beta + g\alpha + b\beta^2 + h\alpha\beta + f\beta + g\alpha + f\beta + c = 0$

$\Rightarrow \alpha(a\alpha + h\beta + g) + \beta(h\alpha + b\beta + f) + g\alpha + f\beta + c = 0$

$\Rightarrow \alpha.0 + \beta.0 + g\alpha + f\beta + c = 0$

$\therefore g\alpha + f\beta + c = 0$……(5)

General Equation of Second degree (Class work)

From (3), (4) and (5) we get

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

$$\therefore \Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

Point of intersection: $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

$$\Rightarrow \frac{\partial}{\partial x}(ax^2 + 2hxy + by^2 + 2gx + 2fy + c) = \frac{\partial}{\partial x}(0)$$

$$\Rightarrow 2ax + 2hy + 2g = 0$$

$$\therefore ax + hy + g = 0 \ldots\ldots\ldots\ldots\ldots (6)$$

Again $\frac{\partial}{\partial y}(ax^2 + 2hxy + by^2 + 2gx + 2fy + c) = \frac{\partial}{\partial y}(0)$

$$\Rightarrow 2hx + 2by + 2f = 0$$

$$\therefore hx + by + f = 0 \ldots\ldots\ldots\ldots\ldots\ldots (7)$$

Solving (6) and (7) we get

$$x = \frac{hf - bg}{ab - h^2} \text{ and } y = \frac{hg - af}{ab - h^2}$$

General Equation of Second degree (Class work)

The general equation of second degree $\boldsymbol{ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0}$ will represent

1 (i) a pair of straight lines if $\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$

(ii) Two parallel lines if $\Delta = 0, ab = h^2$ (no finite point of intersection)

(iii) Two perpendicular lines if $\Delta = 0, a + b = 0$ (From the angle between two curves)

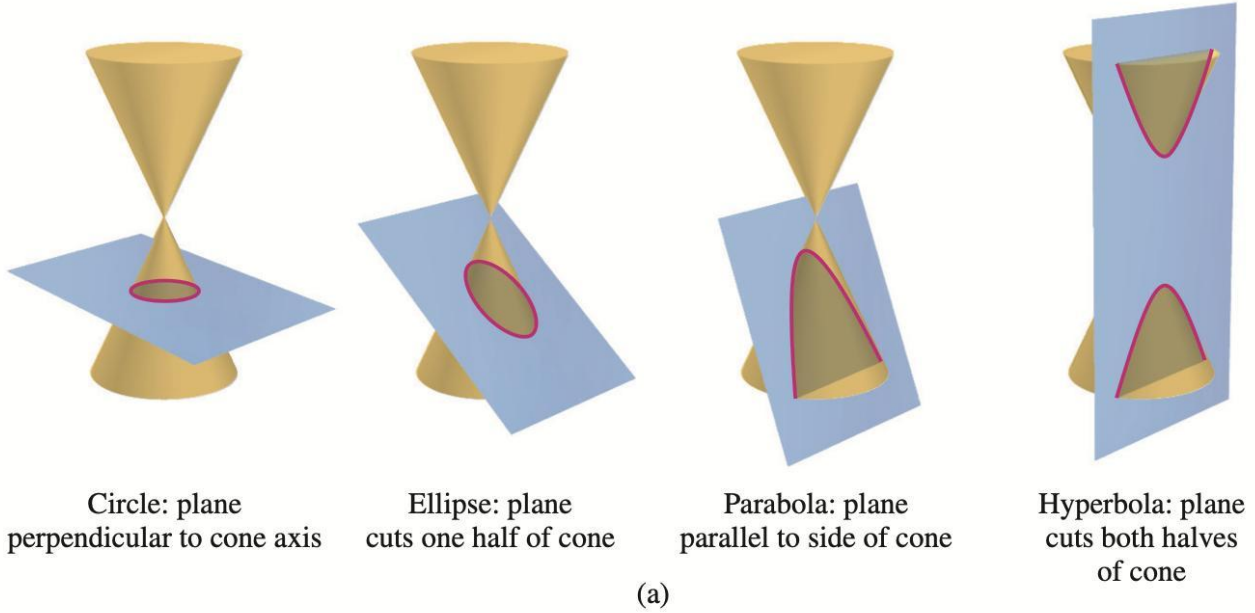
2. a circle if $a = b \neq 0, h = 0$

3. a parabola if $ab = h^2, \Delta \neq 0$

4. an ellipse if $ab - h^2 > 0, \Delta \neq 0$

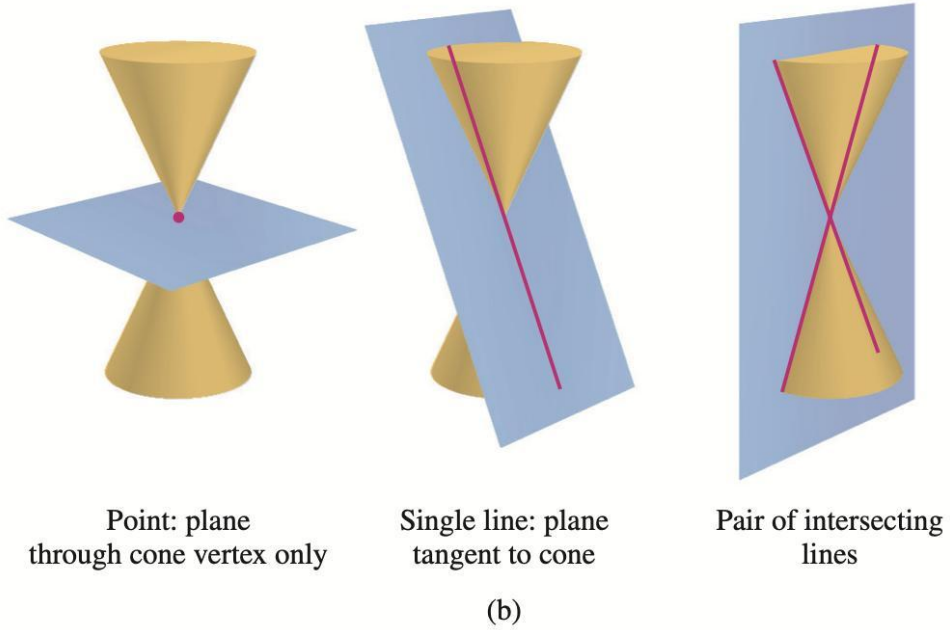
5. a hyperbola if $ab - h^2 < 0, \Delta \neq 0$.

Graphical Representation



A **cone** is a three-dimensional surface. For example, $x^2 + y^2 = z^2$ is a double cone: it has an upper and a lower part meeting at the vertex (0,0,0).

A **conic** is a two-dimensional curve. It is formed when a plane cuts through a cone.



Exercise

What does the equation $5x^2$-2xy+$5y^2$-8x-8y-8=0 represent?

Solution:

The given equation $5x^2$-2xy+$5y^2$-8x-8y-8=0 ……(1)

Comparing (1) with $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

We get a=5, h=-1,b=5,g=-4,f=-4,c=-8

$$\therefore \Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$$

$= abc + 2fgh - af^2$-bg^2-ch^2

=-200-32-80-80+8

=-384

$\therefore \Delta \neq 0$

$ab - h^2 = 5.5 - (-1)^2$=25-1=24

$ab - h^2 > 0$.

Thus the given equation represents an ellipse.

(HW) Prove that the equation $x^2 + 6xy + 9y^2 + 4x + 12y - 5 = 0$ represent a pair of parallel lines. Find equation of straight lines.

Solution: Given $x^2 + 6xy + 9y^2 + 4x + 12y - 5 = 0$

Now compare with equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ then we get

$a = 1, h = 3, b = 9, g = 2, f = 6$ and $c = -5$

$\Delta = abc + 2fgh - af^2$-$bg^2$-c$h^2 = 0$

We have $\Delta = 0$

$$\tan\theta = \frac{2\sqrt{h^2-ab}}{a+b} => \theta = 0$$

Now $x^2 + 6xy + 9y^2 + 4x + 12y - 5 = 0$

$=> x^2 + x(6y + 4) + 9y^2 + 12y - 5 = 0$

$$=> x = \frac{-(6y+4) \pm \sqrt{(6y+4)^2 - 4(9y^2+12y-5)}}{2.1}$$

$\therefore x + 3y + 5 = 0$ and $x + 3y - 1 = 0$

Exercise (HW)

Prove that $3y^2 - 8xy - 3x^2 - 29x + 3y - 18 = 0$ represent a pair of straight. Find also their point of intersection and angles between them.

Solution: Given $3y^2 - 8xy - 3x^2 - 29x + 3y - 18 = 0$

Now compare with general equation of 2nd degree $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ then we get

$a = -3, h = -4, b = 3, g = -\frac{29}{2}, f = \frac{3}{2}$ and $c = -18$

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

We have $\Delta = 0$

$$\tan\theta = \frac{2\sqrt{h^2-ab}}{a+b} => \theta = 90^0$$

$$x = \frac{hf-bg}{ab-h^2} \text{ and } y = \frac{hg-af}{ab-h^2}$$

After solving we get

$x = 3/2$ and $y = 5/2$

So the required point is $(x, y) = (\frac{3}{2}, \frac{5}{2})$

Thanks for your attention